

Nicola Muca Cirone

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EDUCATION	PhD Mathematics - CDT Mathematics of Random Systems, 2022-2025 Imperial College London - University of Oxford • Deep Learning, Mathematical Finance, Rough Paths Analysis. • Supervised by <i>Dr. Cristopher Salvi</i>. • Researching scaling limits of Neural Networks • Working in theoretical foundations for State Space Models. • Developing new Kernel Methods for Time Series analysis. MSc Mathematics, University of Pisa - ETH Zürich 2021-2022 • Honour's degree (110/110 cum Laude) GPA 30 cum Laude / 30 • Stochastic Calculus, Stochastic PDEs, Mathematical Finance, ML in Finance. • Thesis: <i>Kernel Methods on Path Spaces</i> supervised by <i>Prof. Josef Teichmann</i>. BSc Mathematics, University of Pisa 2018-2021 • Honour's degree (110/110 cum Laude) GPA 29.88/30 • Thesis on advanced topics in Geometrical Model Theory.
REVIEWED PUBLICATIONS	[2] Nicola Muca Cirone, Antonio Orvieto, Benjamin Walker, Cristopher Salvi and Terry Lyons. Theoretical Foundations of Deep Selective State-Space Models. <i>Accepted to NeurIPS 2024</i> url: https://arxiv.org/abs/2402.19047 [1] Nicola Muca Cirone, Maud Lemerancier and Cristopher Salvi. Neural signature kernels as infinite-width-depth-limits of controlled ResNets. <i>Proceedings of the 40th International Conference on Machine Learning</i>, 25358–25425, 2023. url: https://proceedings.mlr.press/v202/muca-cirone23a.html
PREPRINTS	[7] Nicola Muca Cirone, Owen Futter and Blanka Horvath. Kernel Learning for Mean-Variance Trading Strategies <i>To appear on Arxiv.</i> [6] Benjamin Walker, Lingyi Yang, Nicola Muca Cirone, Cristopher Salvi and Terry Lyons. Structured Linear CDEs: Maximally Expressive and Parallel-in-Time Sequence Models. url: https://arxiv.org/abs/2505.17761 [5] Sajad Movahedi, Felix Sarnthein, Nicola Muca Cirone and Antonio Orvieto. Fixed-Point RNNs: From Diagonal to Dense in a Few Iterations url: https://arxiv.org/abs/2503.10799 [4] Nicola Muca Cirone and Cristopher Salvi. ParallelFlow: Parallelizing Linear Transformers via Flow Discretization. url: https://arxiv.org/abs/2504.00492 [3] Mie Glückstad, Nicola Muca Cirone and Josef Teichmann. Signature Reconstruction from Randomized Signatures. <i>Submitted to Journal</i> url: https://arxiv.org/abs/2502.03163

- [2] **Nicola Muca Cirone** and Cristopher Salvi.
Rough kernel hedging.
url: <https://arxiv.org/abs/2501.09683>
- [1] **Nicola Muca Cirone**, Jad Hamdan and Cristopher Salvi.
Graph Expansions of Deep Neural Networks and their Universal Scaling Limits.
Submitted to Journal
url: <https://arxiv.org/abs/2407.08459>

EXPERIENCE	Intern - ML Researcher , Cartesia AI San Francisco	Feb - May 2025
	Building new ways to distill LLMs to different architectures, depths and sizes. Research on development of ViT models capable of multi-modal reasoning.	
	Industry Project - Neural Signature Kernels , Imperial College London	June 2024
	Co-organized 2024 CDT Industry Project and supervised PhD students in applying Neural Signature Kernels to time series analysis.	
	Teaching Assistant - Stochastic Calculus for Finance , ICL Business School <i>MSc Risk Management & Financial Engineering</i> . Taught exercise classes.	Oct - Dec 2023
	Industry Project - Brain signal analysis with Signatures , Imperial College London Challenge: analyze MEG data from several subjects, detect the one who had a seizure. Our team used Signature techniques, the success and clarity of our results won the competition.	June 2023
TALKS	Teaching Assistant - Numerical Methods for Finance , Imperial College London <i>MSc Mathematics and Finance</i> . Taught exercise classes.	Feb 2023
	Teaching Assistant - Data Analysis , Università di Pisa <i>MSc Mathematics</i> . Provided tutoring support.	2021-2022
	Workshop on Recent Developments in Theoretical Machine Learning , Imperial-X Title: <i>Graph Expansions of Deep Neural Networks</i> .	Jan 2025
	Stochastic Analysis Seminar , ShanghaiTech University Title: <i>Theoretical Foundations of Deep Selective State-Space Models</i> .	Nov 2024
	DataSig - Rough Paths Interest Group , Alan Turing Institute Title: <i>Signature Reconstruction from Randomized Signatures</i> .	May 2024
	Computer Vision Group Seminar , University of Michigan Title: <i>Theoretical Foundations of Deep Selective State-Space Models</i> .	April 2024
	Talks in Financial and Insurance Mathematics , ETH Zürich Title: <i>Neural Signature Kernels (and Trees)</i> .	Dec 2023
	7th London-Paris Bachelier Workshop , Imperial College London Title: <i>Rough Kernel Hedging</i> .	Sept 2023
	CDT Mathematics of Random Systems Workshop , University of Oxford Title: <i>Neural Signature Kernels</i> .	June 2023
	16th Oxford-Berlin Young Researchers Meeting , University of Oxford Title: <i>Signatures and the infinite-width-depth limit of Data Driven ResNets</i> .	Dec 2022
AWARDS	NeurIPS 2024 Scholar Award , Neural Information Processing Systems Foundation	2024
	INDAM Scholarship , Istituto Nazionale di Alta Matematica Two year scholarship from the Italian Institute of Higher Mathematics. Placed 2nd at National level.	2022
COMPETENCES	Languages Italian (<i>native</i>), English (C1), French (<i>basic</i>)	
	Technical Skills Python, Jax, Pytorch, Triton, CUDA, R, Matlab, Java	